

PROBLEM STATEMENT

Modern drug discovery is slow, expensive, and highly inefficient, often taking 10 to 15 years and billions of dollars to develop a single drug, with most candidates failing during clinical trials.

At the same time, many existing approved drugs may have untapped potential to treat other diseases, but identifying these opportunities requires analyzing vast amounts of fragmented biomedical data involving genes, proteins, diseases, and drug interactions.

Researchers currently lack efficient tools to uncover these hidden relationships, leading to missed therapeutic opportunities and delayed treatment development for patients.

PAIN POINTS

- Drug discovery is extremely time-consuming, often taking 10–15 years to bring a new drug to market.
- Drug development is highly expensive, with costs frequently exceeding \$1 billion per approved drug.
- Most drug candidates fail during clinical trials, resulting in significant wasted time and resources.
- Biomedical data is scattered across multiple databases and research publications, making comprehensive analysis difficult.
- Potential new uses for existing drugs (drug repurposing opportunities) are often overlooked due to the complexity of analyzing disease, protein, gene, and drug relationships.

TOP EXISTING SOLUTIONS AND GAPS

Existing Competitor	What They Do	Key Gap / Limitation
BenevolentAI	Identifies dysregulated biological pathways and predicts novel drug targets using AI.	Focuses primarily on target discovery rather than providing a complete drug repurposing and researcher exploration platform.
Chai Discovery	Uses AI for antibody and protein design, with a strong focus on antibody engineering.	Limited to antibody-centric applications and does not broadly address drug repurposing across all drug categories.
Isomorphic Labs	Uses AlphaFold based protein modeling and AI to design and optimize new drug candidates.	Focuses on creating new drugs rather than identifying new uses for existing approved drugs.
Healx	Uses AI driven drug repurposing to identify treatments for rare diseases.	Primarily focused on rare diseases, limiting its applicability across a wider range of therapeutic areas.